

1. Introduction

In Week 2, the cleaned **Superstore Sales dataset** from Week 1 was used to create a structured data model and an interactive dashboard using **Microsoft Power BI**.

The main focus of this project was to create dimension tables, establish relationships, develop DAX measures, create KPI cards and visualizations, and add interactive filtering features.

2. Objectives

The objectives of Week 2 were:

- To create a structured data model using Power BI.
- To create dimension tables for Date, Customer, Product and Region.
- To establish relationships between tables.
- To create DAX measures for important business metrics.
- To create KPI cards for sales and profitability analysis.
- To create charts for category, sub-category, regional and time-based analysis.
- To add interactive slicers and cross-filtering.
- To develop an interactive Superstore Sales Dashboard.

3. Dataset Used

The dataset used for this project is the **Superstore Sales Dataset**.

The main table in the Power BI file is named:

samplesuperstore

The table contains sales-related information such as:

- Order Date
- Order ID
- Customer ID
- Customer Name
- Product ID
- Product Name
- Category
- Sub-Category

- Region
- Sales
- Profit
- Quantity
- Segment

The cleaned dataset prepared during Week 1 was used for the Week 2 analysis.

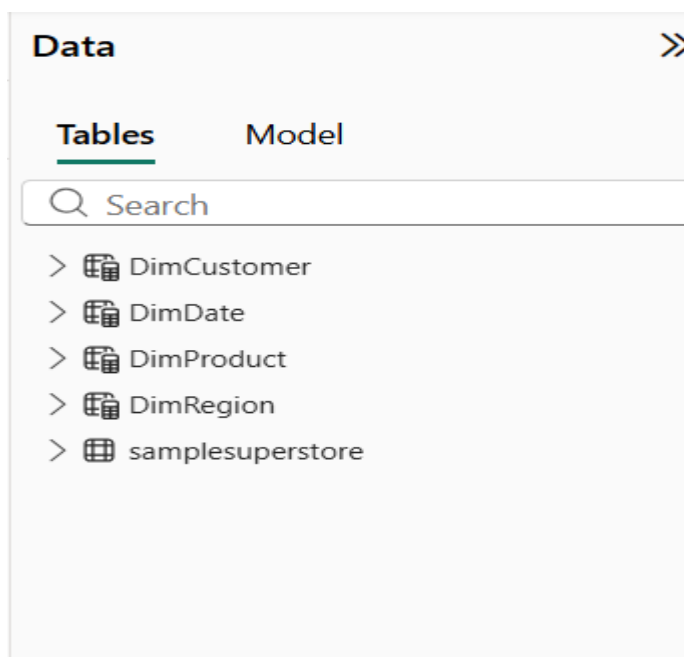
4. Data Modeling

A structured data model was created using the main samplesuperstore table and supporting dimension tables.

The following tables were created:

- samplesuperstore
- DimDate
- DimCustomer
- DimProduct
- DimRegion
- _Measures

The samplesuperstore table was used as the central transaction table.



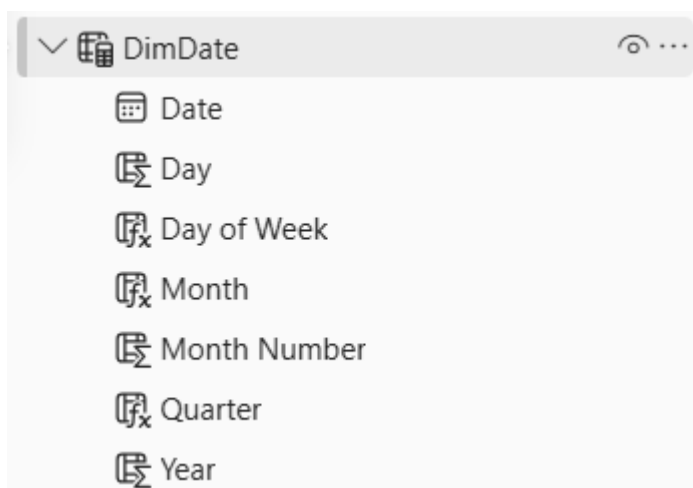
5. DimDate Table

A separate **DimDate** table was created using the Order Date field.

The following date attributes were created:

- Date
- Day
- Day of Week
- Month
- Month Number
- Quarter
- Year

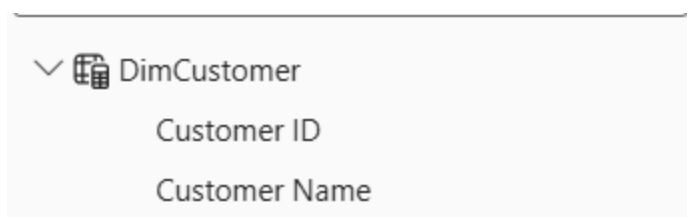
These fields are useful for analyzing sales performance over different time periods.



6. Customer Dimension

A separate **DimCustomer** table was created using Customer ID and Customer Name from the samplesuperstore table.

The table contains unique customer information.



7. Product and Region Dimensions

Two additional dimension tables were created:

DimProduct

Contains:

- Product ID
- Product Name
- Category
- Sub-Category

DimRegion

Contains:

- Region

These tables help analyze product and regional performance.



8. Table Relationships

Relationships were established between the dimension tables and the main samplesuperstore table.

The main relationships were:

Dimension Table	Main Table
DimDate[Date]	samplesuperstore[Order Date]
DimCustomer[Customer ID]	samplesuperstore[Customer ID]
DimProduct[Product ID]	samplesuperstore[Product ID]

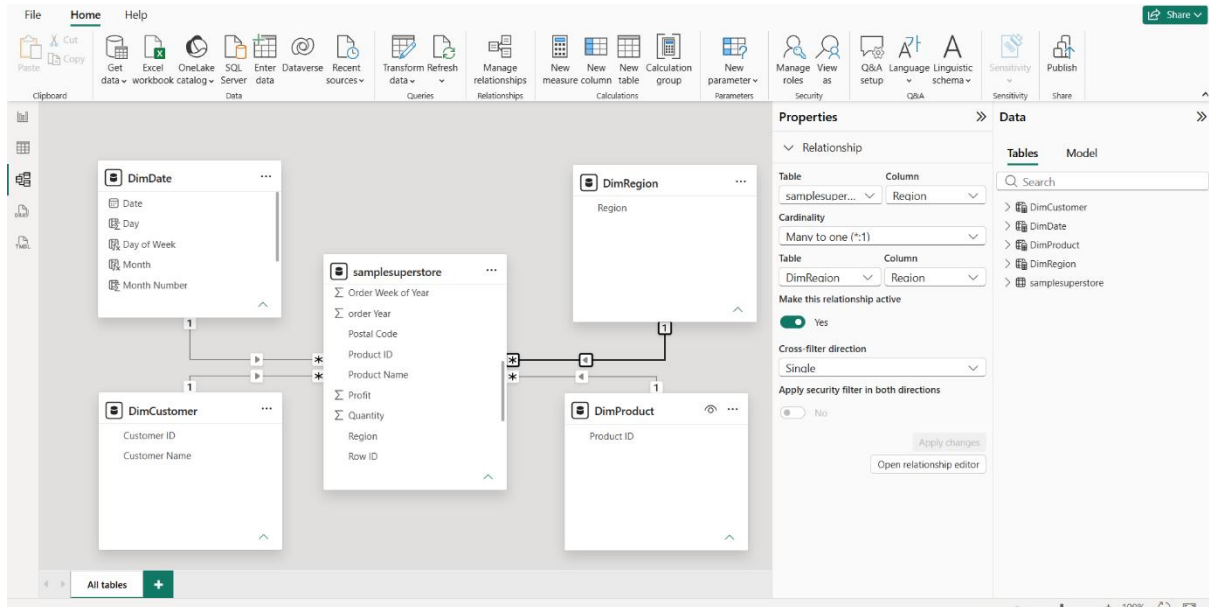
Dimension Table

Main Table

DimRegion[Region]

samplesuperstore[Region]

The relationships were created using a **one-to-many (1:*)** structure.



9. DAX Measures

A dedicated _Measures table was created to store the DAX calculations used in the dashboard.

The following measures were created.

Total Sales

Total Sales =

SUM('samplesuperstore'[Sales])

Total Profit

Total Profit =

SUM('samplesuperstore'[Profit])

Total Orders

Total Orders =

DISTINCTCOUNT('samplesuperstore'[Order ID])

Total Quantity

Total Quantity =

SUM('samplesuperstore'[Quantity])

Average Sales

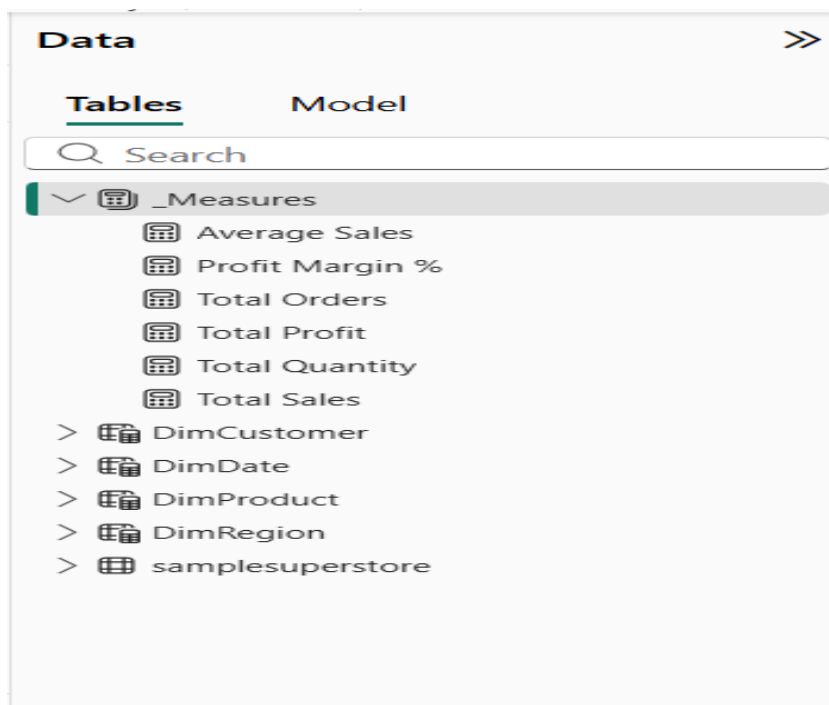
Average Sales =

AVERAGE('samplesuperstore'[Sales])

Profit Margin %

Profit Margin % =

DIVIDE(
[Total Profit],
[Total Sales],
0
)



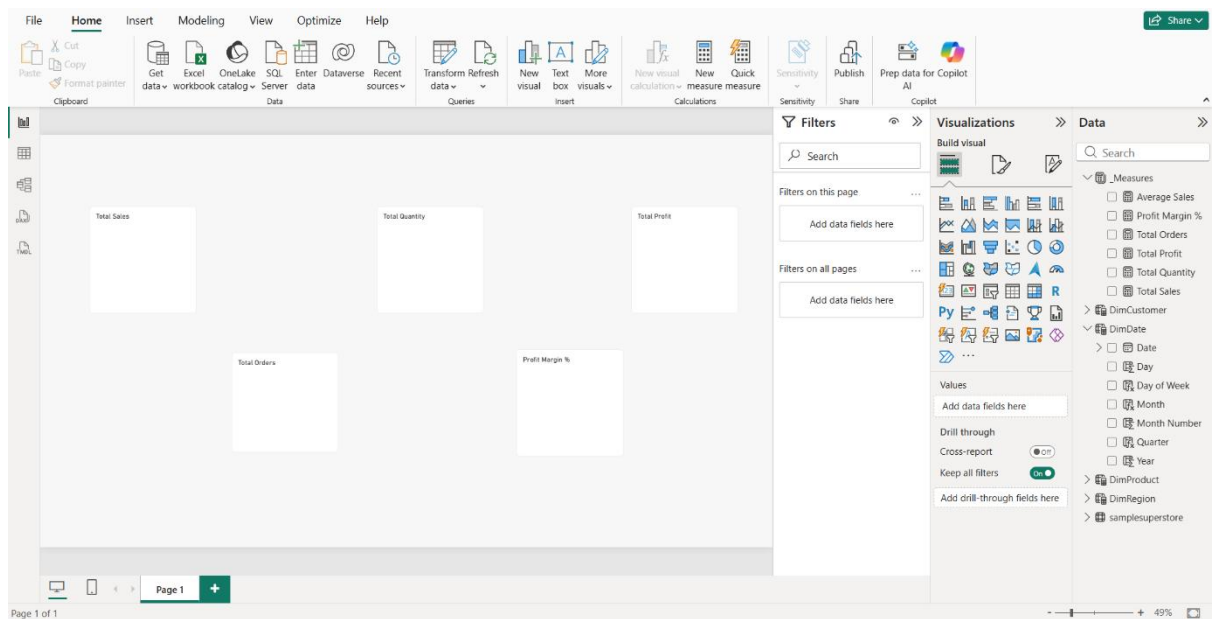
10. KPI Cards

KPI cards were created to display important business performance indicators.

The following KPIs were included:

- Total Sales
- Total Profit
- Total Orders

- Total Quantity
- Profit Margin %

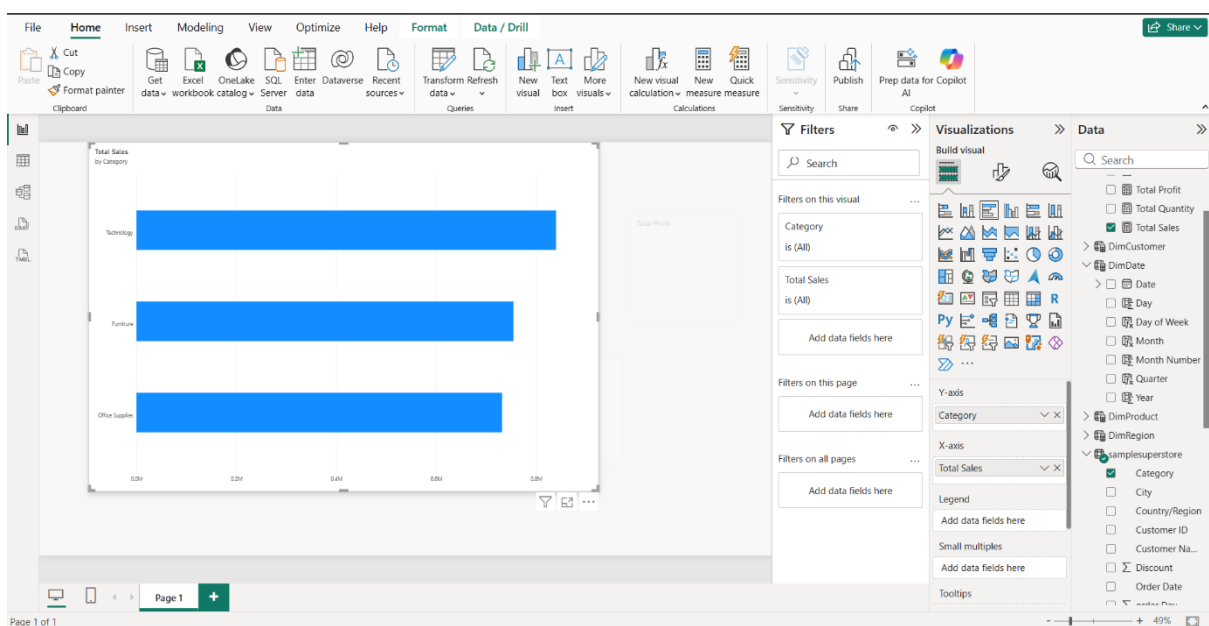


11. Sales by Category

A bar chart was created to compare sales across different product categories.

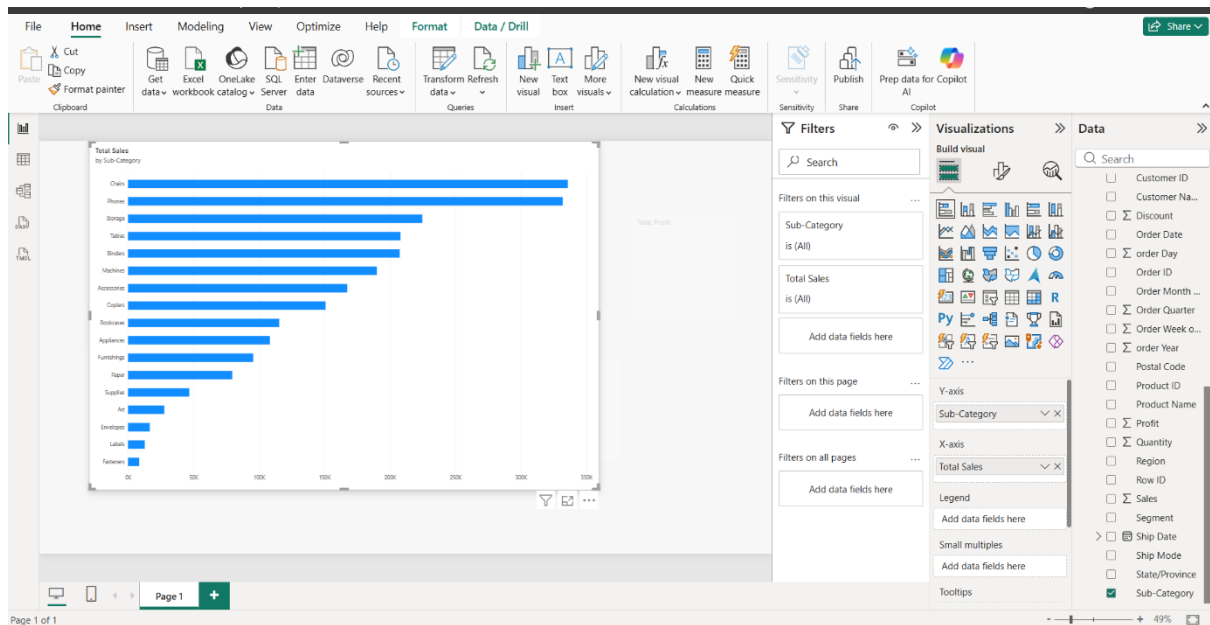
The visual uses:

- **Category** as the category field
- **Total Sales** as the value



12. Sales by Sub-Category

A bar chart was created to provide a more detailed analysis of sales by sub-category.

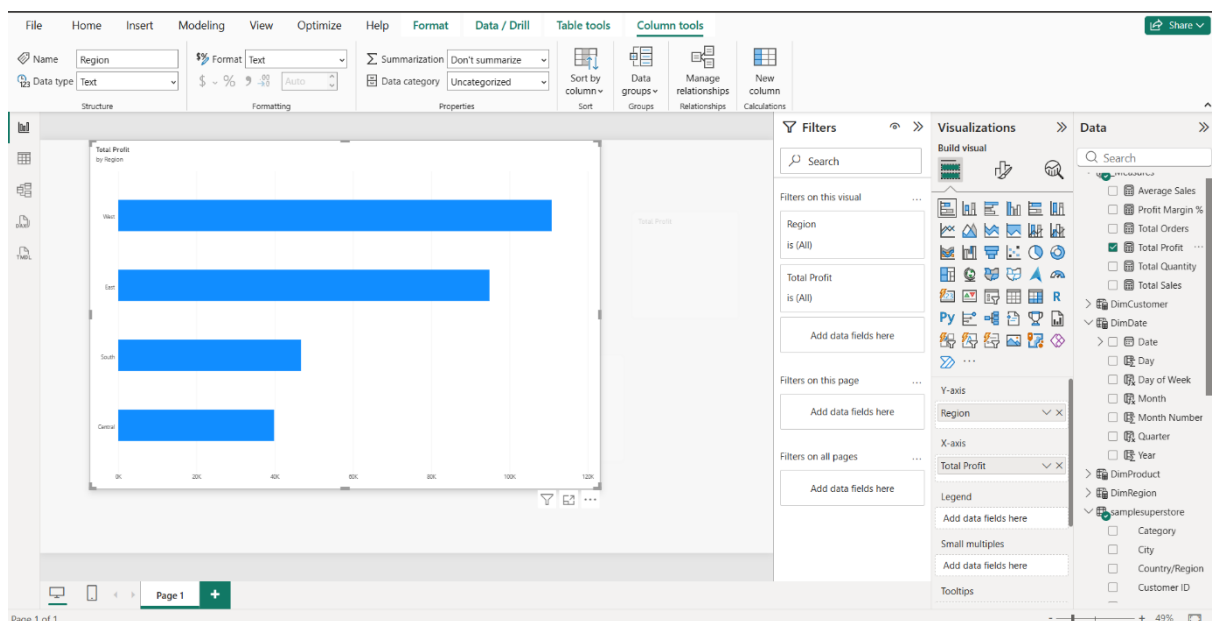


13. Profit by Region

A bar chart was created to compare total profit across different regions.

The visual uses:

- **Region**
- **Total Profit**

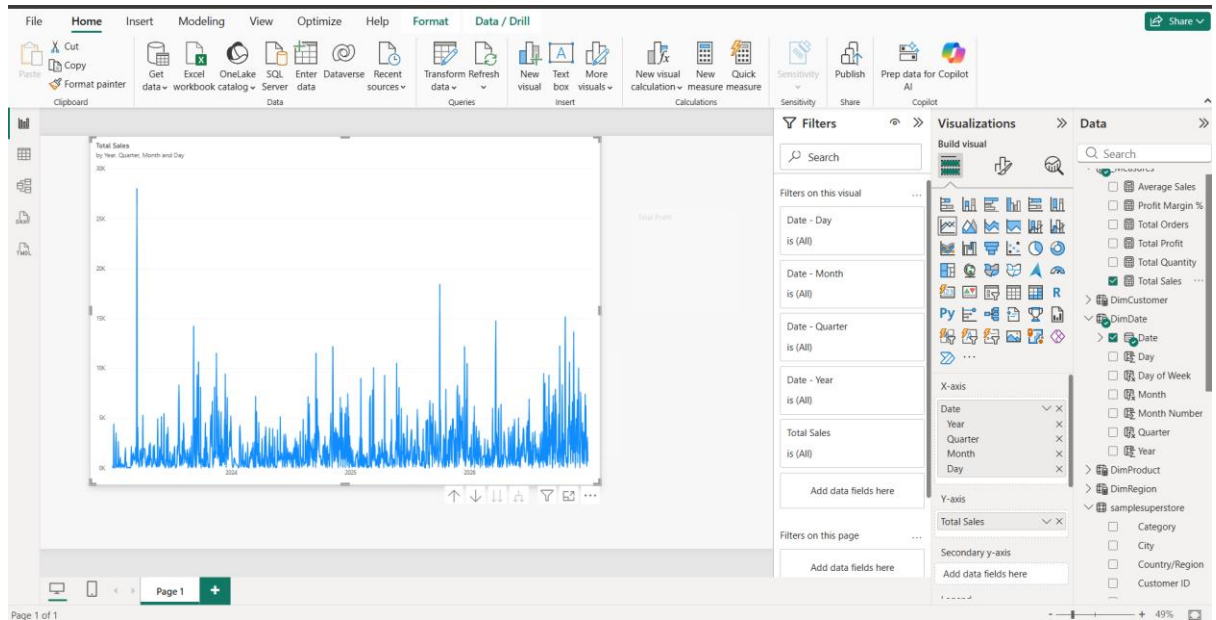


14. Sales Trend Over Time

A line chart was created to analyze changes in sales over time.

The visual uses:

- **Date** from DimDate
- **Total Sales**



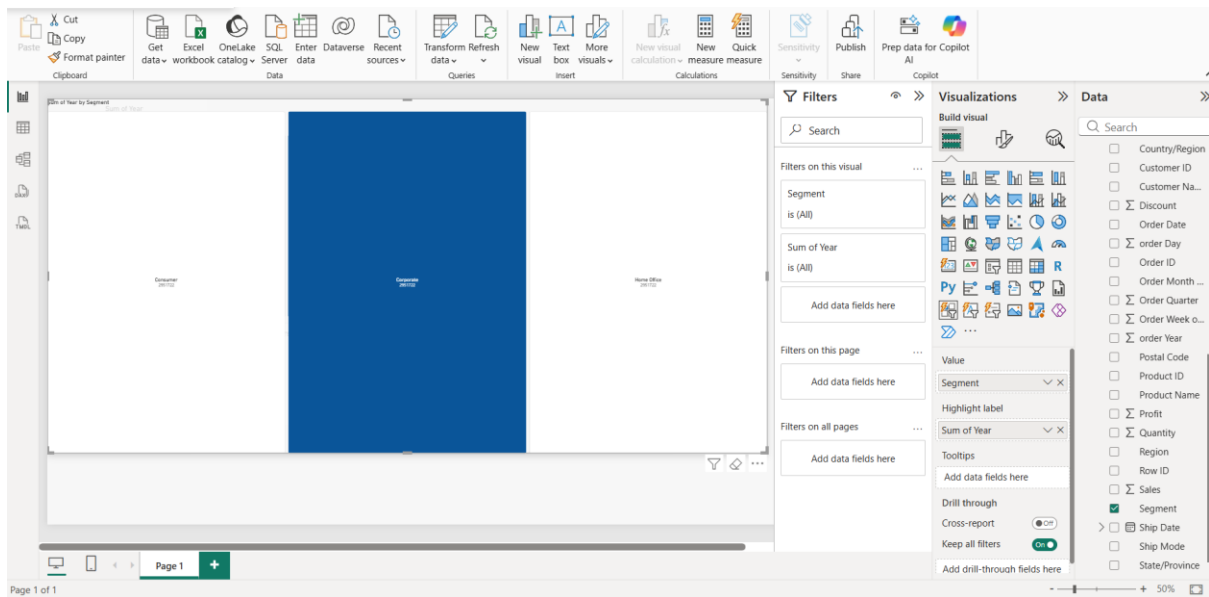
15. Interactive Slicers

Slicers were added to make the dashboard interactive.

The following slicers were created:

- Year
- Region
- Category
- Segment

Users can select different values from these slicers to filter the dashboard



16. Conclusion

In Week 2, the cleaned Superstore Sales dataset was successfully transformed into a structured Power BI data model. Dimension tables were created for Date, Customer, Product and Region, and relationships were established with the main samplesuperstore table.

DAX measures were created to calculate Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales and Profit Margin. These measures were used to create KPI cards and different visualizations.

Interactive features such as slicers, cross-filtering and drill-down were also implemented.

Finally, all the visualizations were combined into an interactive **Superstore Sales Dashboard**.

The completed dashboard provides a clear way to analyze sales, profit, orders, product categories and regional performance.